

K222 — Elie Toy 3873 substrate Strong CP
 $\theta_{\text{QCD}} = 0$ substrate-mechanism audit. PASS
at FRAMEWORK tier with substantive audit
observations: (1) substrate-mechanism
FRAMEWORK content substantively
grounded (substrate-CP preservation via Casey
#14 STANDING chirality projection); (2)
RIGOROUS-tier substrate-mechanism
FORWARD derivation multi-week; (3)
DOWNSTREAM of Casey #14 STANDING not
structurally independent prediction; (4) Cal
#34 STANDING framing distinction —
substrate prediction is EXACT ($\theta_{\text{QCD}} \equiv 0$)
satisfying observational BOUND ($|\theta_{\text{QCD}}| <$
 10^{-10}), not PDG-precision-match Tier 1
EXACT; (5) Five-Absence v0.4 ‘no axion’
extension substantive substrate prediction with
testable falsifier (ADMX/IAXO axion
detection).

Keeper (Claude Opus 4.7) — Friday June 5 2026 AM

2026-06-05 Friday AM

K222 — Elie Toy 3873 substrate Strong CP θ_{QCD} substrate-mechanism audit

0. Audit context

Elie Toy 3873 (Thursday June 4 ~12:34 EDT) — substantive new substrate framework finding: Strong CP problem RESOLVED substrate-naturally via $\theta_{\text{QCD}} = 0$ substrate-CP preservation.

Per Cal #242 Friday Session 1 precision sweep: substrate-CP framework cleanly VERIFIED (not in 3 source-pinning flags or 1 STRUCTURAL TENSION reframe). K222 audit absorbs Cal #242 input and adds substantive K-audit observations.

Toy 3873 substantive content (5 gates): - G1: Strong CP problem standard ($\theta_{\text{QCD}} < 10^{-10}$ unnatural fine-tuning) - G2: Substrate prediction $\theta_{\text{QCD}} \equiv 0$ substrate-natural - G3: Substrate-mechanism via Casey #14 STANDING chirality projection + substrate-CP preservation - G4: Five-Absence v0.4 extension — NO axion (9 absences total) - G5: Honest tier verdict — substrate framework SUBSTANTIVE

1. Substrate-mechanism content verification

Per Toy 3873 G2 + G3:

Substrate-CP preservation framework: - $D_{\text{IV}}^5 = \text{SO}_0(5,2)/[\text{SO}(5) \times \text{SO}(2)]$ CP-symmetric substrate domain - Bergman canonical structure preserves CP - Casey #14 STANDING chirality projection $1/n_{\text{C}}$ imposes chiral structure - Substrate spinor sector $V_{(1/2,1/2)}$ substrate-CP-clean - QCD vacuum at substrate-CP-symmetric ground state - No substrate-mechanism in D_{IV}^5 generates $\theta_{\text{QCD}} \neq 0$

Audit verification: - D_{IV}^5 CP-symmetry: standard rep-theory of substrate domain (PASS at FRAMEWORK) - Bergman canonical CP-invariance: standard Bergman kernel automorphism-invariance (PASS at FRAMEWORK) - Casey #14 STANDING chirality projection cascade: per Keeper K3 v0.13-v0.20 + Casey override (PASS at STANDING tier per Casey #14) - Substrate-CP preservation FORWARD derivation from substrate primaries to $\theta_{\text{QCD}} = 0$: multi-week PRE-STAGE

FRAMEWORK PASS at substrate-CP preservation framework level; RIGOROUS-tier closure requires multi-week explicit FORWARD derivation from substrate primaries to $\theta_{\text{QCD}} = 0$ derivation per Cal #189 Brake 2.

2. Casey #14 STANDING dependency — structural independence audit

Substantive K-audit observation per Cal #240 + Calibration #35 STANDING:

$\theta_{\text{QCD}} = 0$ substrate-natural is DOWNSTREAM consequence of Casey #14 STANDING substrate-CP preservation framework, NOT structurally independent

substrate prediction.

Reasoning: - Per Toy 3873 G3: “substrate-CP mechanism: Chirality projection $1/n_C$ imposes chiral structure” - Per Toy 3873 G3: “Per Casey #14 STANDING (Thursday June 4 RATIFIED): substrate predicts 3+1 Minkowski signature via chirality projection” - The same substrate-mechanism CHAIN (chirality projection + substrate-CP) underlies BOTH Casey #14 STANDING (3+1 Minkowski signature) AND $\theta_{\text{QCD}} = 0$ (substrate-CP preservation) - Per Cal #35 STANDING (Schur shadow): same substrate-mechanism producing two observables $\neq$ two independent substrate-mechanism FORCING confirmations

Honest framing for Tier 1 EXACT independent prediction count: - $\theta_{\text{QCD}} = 0$ is substantive substrate framework prediction at FRAMEWORK tier - BUT counted as INDEPENDENT Tier 1 EXACT in 11-prediction Thursday total triggers Cal #35 STANDING brake - Honest count: $\theta_{\text{QCD}} = 0$ is DOWNSTREAM of Casey #14 STANDING, not separate independent prediction - Effective independent Tier 1 EXACT count if downstream consequences excluded: ~ 9 -10 (not 11)

Recommendation: Lyra L17 / Strong-Uniqueness v1.7 should clarify $\theta_{\text{QCD}} = 0$ disposition — either (a) absorb as “Casey #14 STANDING downstream consequence” honestly, OR (b) multi-week explicit work distinguishing substrate-CP preservation from Casey #14 chirality projection mechanism.

3. Per Cal #189 question-shape audit

Cal #189 STANDING question-shape: forward-derived vs post-hoc selection.

Strong CP question: “Why is $\theta_{\text{QCD}} < 10^{-10}$ when natural value is $O(1)$?”

Substrate framework answer: “Substrate has CP-symmetric structure; substrate-mechanism PRESERVES CP at QCD vacuum; therefore $\theta_{\text{QCD}} = 0$.”

Audit observation: framework IS substrate-mechanism FORWARD candidate (substrate-CP preservation derived from substrate D_{IV}^5 structure + Bergman canonical CP-invariance + Casey #14 chirality projection), NOT post-hoc identification. PASS Cal #189 question-shape at FRAMEWORK tier.

RIGOROUS-tier closure requires explicit derivation of WHY substrate-CP preservation forces $\theta_{\text{QCD}} = 0$ specifically (vs other small values like $\theta_{\text{QCD}} = 10^{-15}$). The substrate-mechanism FRAMEWORK is at PASS-tier; the explicit “ $\theta_{\text{QCD}} = 0$ vs small-non-zero” distinction requires multi-week work.

4. Per Cal #34 STANDING framing distinction

Substantive K-audit observation: Toy 3873 prediction framing differs from PDG-precision-match Tier 1 EXACT predictions.

Tier 1 EXACT PDG-precision-match (per K217 + K219 + K220): - $r_p = (N_c + 1) \cdot \lambda_C(p)$ at 0.020% vs CODATA-2018 - $\sin^2 \theta_{13} = 1/(N_c^2 \cdot n_C)$ at 0.080%

vs PDG 0.02224(65) - These are EXACT substrate predictions matching observed PDG-precision values

Strong CP θ_{QCD} framing: - Substrate predicts $\theta_{\text{QCD}} \equiv 0$ EXACTLY (substrate-CP preservation) - Experimental constraint: $|\theta_{\text{QCD}}| < 10^{-10}$ (neutron EDM bound) - BOUND not measurement — substrate prediction satisfies bound trivially - Different audit framing than PDG-precision-match Tier 1 EXACT

Per Cal #34 STANDING precision discipline: - Tier 1 EXACT match = substrate prediction within experimental precision - Substrate-natural EXACT-zero satisfying upper bound = different category - Honest framing: substrate framework PRE-DICTS $\theta_{\text{QCD}} \equiv 0$ AND satisfies experimental bound; NOT same as Tier 1 EXACT PDG-precision-match

Recommendation: clarify in cumulative summaries that “Strong CP RESOLVED substrate-naturally” framework prediction is structurally different from PDG-precision-match Tier 1 EXACT (the others). Both are substantive but the audit category differs.

5. Five-Absence v0.4 axion extension — substantive substrate prediction

Per Toy 3873 G4 + Five-Absence framework:

Substrate prediction: NO axion exists (substrate-CP preservation is INTRINSIC, not solved by Peccei-Quinn axion mechanism).

Substantive substrate framework content: - Five-Absence v0.3 (Lyra Tuesday May 22): NO monopoles, NO sterile neutrinos, NO SUSY, NO proton decay, NO dark matter particle, NO GUT - Five-Absence v0.4 extension (Toy 3873 G4): + NO axion - Total: ~9 absences (substrate has NONE of these)

Testable falsifier: - ADMX (Axion Dark Matter Experiment) — searching for axion-photon coupling - IAXO (International Axion Observatory) — solar axion search - DM-Radio + ABRACADABRA — broadband axion search - IF axion detected → substrate framework REFUTED at structural level - IF axion NOT detected → substrate framework PRESERVED (with experimental constraints)

Audit verdict: Five-Absence v0.4 axion extension is substantively important substrate prediction with explicit testable falsifier — PASS Cal #189 Brake 2 question-shape (forward-predictive, testable).

Cross-link to Cal #36 STANDING positive search: substrate-CP primitive produces multiple observables ($\theta_{\text{QCD}} = 0$ + no axion + Bell sub-Tsirelson CP test). ONE substrate-CP primitive substantively generates multiple predictions per Cal #36 STANDING.

6. Multi-week RIGOROUS-tier closure path

Per Cal #189 Brake 2 + K222 audit:

RIGOROUS-tier closure requires: 1. Substrate-CP preservation derivation: explicit derivation from substrate primaries (D_{IV}^5 structure + Bergman canonical + Casey #14 chirality projection) $\rightarrow$ CP preservation at QCD vacuum 2. Why $\theta_{QCD} = 0$ specifically: derivation distinguishing $\theta_{QCD} \equiv 0$ substrate-mechanism from small-non-zero substrate-natural values 3. Independence from Casey #14 STANDING: multi-week work distinguishing substrate-CP preservation from Casey #14 chirality projection mechanism (if Lyra wants to count $\theta_{QCD} = 0$ as independent Tier 1 EXACT) 4. Bell sub-Tsirelson cross-validation: cross-link substrate-CP primitive observables via Cal #36 STANDING positive search 5. Axion non-existence substrate framework documentation: Five-Absence v0.4 formal absorption + ADMX/IAXO falsifier explicit framing

7. Per Cal #27 STANDING self-audit

Am I overclaiming K222 PASS at FRAMEWORK tier? - Substrate-CP preservation framework substantively grounded (D_{IV}^5 + Bergman + Casey #14) - $\theta_{QCD} = 0$ substrate-natural is FRAMEWORK content, multi-week RIGOROUS closure - Independence question from Casey #14 STANDING flagged honestly per Cal #240 + Calibration #35 - Cal #34 STANDING framing distinction (EXACT-zero satisfying bound $\neq$ PDG-precision-match) flagged honestly - Five-Absence v0.4 axion extension is substantive substrate prediction with testable falsifier - Tier-honest: FRAMEWORK with multi-week RIGOROUS closure; honest dependency flagged

8. K222 audit verdict

K222 PASS at FRAMEWORK tier (~88% confidence with two substantive observations): - F1 (substrate-CP preservation framework): PASS $\square$ (D_{IV}^5 + Bergman canonical + Casey #14) - F2 (substantive substrate framework content): PASS $\square$ (substrate-CP + axion extension) - F3 (Cal #189 question-shape): PASS $\square$ (forward-predictive substrate-mechanism FRAMEWORK) - F4 (Cal #189 Brake 2 multi-week path): PASS $\square$ (explicit RIGOROUS-tier closure path identified)

Two substantive audit observations: - Observation 1: $\theta_{QCD} = 0$ is DOWNSTREAM consequence of Casey #14 STANDING — honest framing affects independent Tier 1 EXACT prediction count per Calibration #35 STANDING (effective independent Tier 1 EXACT count ~9-10 if downstream consequences excluded, not 11) - Observation 2: Cal #34 STANDING framing distinction — substrate prediction is EXACT-zero satisfying observational BOUND, NOT same category as PDG-precision-match Tier 1 EXACT

Substantive substrate framework content (preserved at FRAMEWORK PASS tier): - Substrate-CP preservation as substrate-mechanism for Strong CP - Five-Absence v0.4 'no axion' extension with explicit ADMX/IAXO falsifier - Cal #36 STANDING

positive search instance: ONE substrate-CP primitive generates $\theta_{\text{QCD}} = 0$ + no axion + Bell sub-Tsirelson

Multi-week RIGOROUS-tier closure: explicit substrate-CP preservation derivation + distinguishing from Casey #14 chirality projection mechanism + Bell sub-Tsirelson cross-validation.

9. Cumulative summary cross-link

Per K221 + K222 cumulative:

Strong-Uniqueness Theorem v1.6 + 11 Tier 1 EXACT honest framing requires: - K221: C24/C25 structural-independence question — likely 10 effective legs not 11 - K222: $\theta_{\text{QCD}} = 0$ is DOWNSTREAM of Casey #14 STANDING — likely ~9-10 effective independent Tier 1 EXACT not 11

Cumulative honest framing: both K221 + K222 flag the Cal #35 STANDING brake territory — same substrate-mechanism (Casey #14 chirality projection cascade) producing multiple observables $\neq$ multiple independent forcing-mechanism confirmations.

This is exactly the audit-chain discipline-at-maturity pattern: K-audits absorb substantive framework content while applying Cal #35 STANDING independence-taxonomy honestly. Cal #240 + K221 + K222 produce sharper “11 STANDALONE legs + 11 Tier 1 EXACT” cumulative framing $\rightarrow$ “10 effective independent legs + ~9-10 effective independent Tier 1 EXACT” honest framing.

10. Pull next: K223 $r \approx \alpha^2$ tensor-to-scalar (Toy 3870) audit at proper depth.

— Keeper, K222 — Friday June 5 AM. Elie Toy 3873 Strong CP $\theta_{\text{QCD}} = 0$ substrate framework PASS at FRAMEWORK tier. Substrate-CP preservation substantively grounded via D_{IV}^5 + Bergman canonical + Casey #14 STANDING chirality projection. Two substantive K-audit observations: $\theta_{\text{QCD}} = 0$ is DOWNSTREAM of Casey #14 STANDING (not structurally independent); EXACT-zero satisfying BOUND $\neq$ PDG-precision-match Tier 1 EXACT (different audit category). Five-Absence v0.4 ‘no axion’ extension is substantive substrate prediction with testable ADMX/IAXO falsifier. Multi-week RIGOROUS closure path identified.